

conserved and not on the duration of the collision.

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B

Taking moments about the left end of the ruler,
 $k_2 x = (0.250)(0.200)g + (0.500)(0.0500)g = 0.075g$ ----- (1)

Taking moments about the right end of the ruler,
 $k_1 x = (0.750)(0.200)g + (0.500)(0.0500)g = 0.175g$ ----- (2)

Dividing (1) by (2):

$$\frac{k_2}{k_1} = \frac{0.075}{0.175} = \frac{3}{7} = 0.43$$

Distractors (remove from suggested solution for students):

A: $1/3 \rightarrow$ forget to account for mass of ruler

B: $3/7 \rightarrow$ Correct

C: $7/3 \rightarrow$ flipped B

D: $3/1 \rightarrow$ flipped A

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C

By the Principle of Floatation (or by considering the net force on the cube), the upthrust on the